

Lab Title: Design and configure a small LAN using Packet Tracer with at least 4 PCs, a switch, and a router. Assign IP addresses manually and test connectivity using ping.

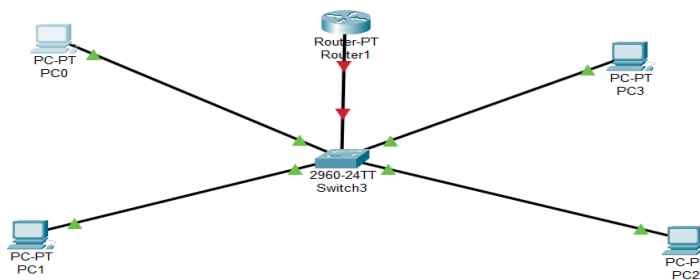
Objective:

To design and configure a small Local Area Network (LAN) consisting of 4 PCs, a switch, and a router. Assign IP addresses manually and test the connectivity using the ping command.

Apparatus/Software Required:

- Cisco Packet Tracer
- 4 PCs
- 1 Switch
- 1 Router
- Copper Straight-Through Cables

Network Topology Diagram:



IP Addressing Scheme:

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet 0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	FastEthernet 0	192.168.10.11	255.255.255.0	192.168.10.1
PC2	FastEthernet 0	192.168.10.12	255.255.255.0	192.168.10.1
PC3	FastEthernet 0	192.168.10.13	255.255.255.0	192.168.10.1
Router	FastEthernet 0/0	192.168.10.1	255.255.255.0	-

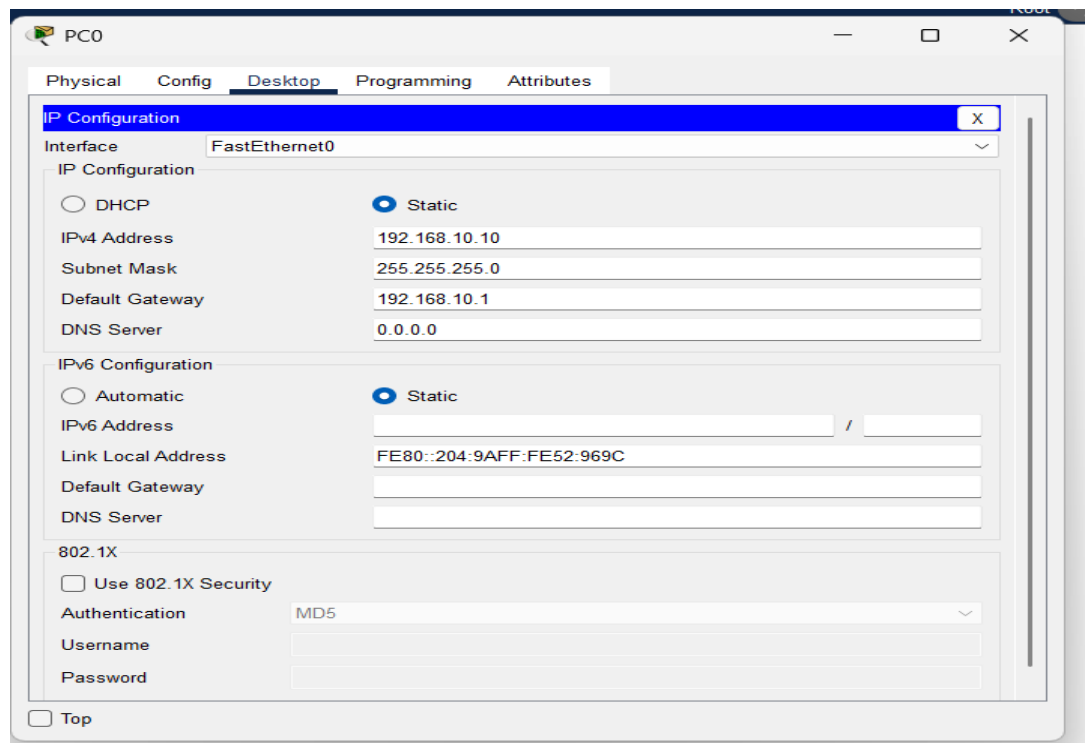
Procedure:

Step 1: Device Placement and Cabling

- Place 4 PCs, a Switch, and a Router in the workspace.
- Use Copper Straight-Through Cables to connect:
 - PCs to Switch
 - Switch to Router

Step 2: IP Address Assignment on PCs

- Click on each PC → Desktop → IP Configuration.
- Enter the IP address, subnet mask, and default gateway as per the table above.

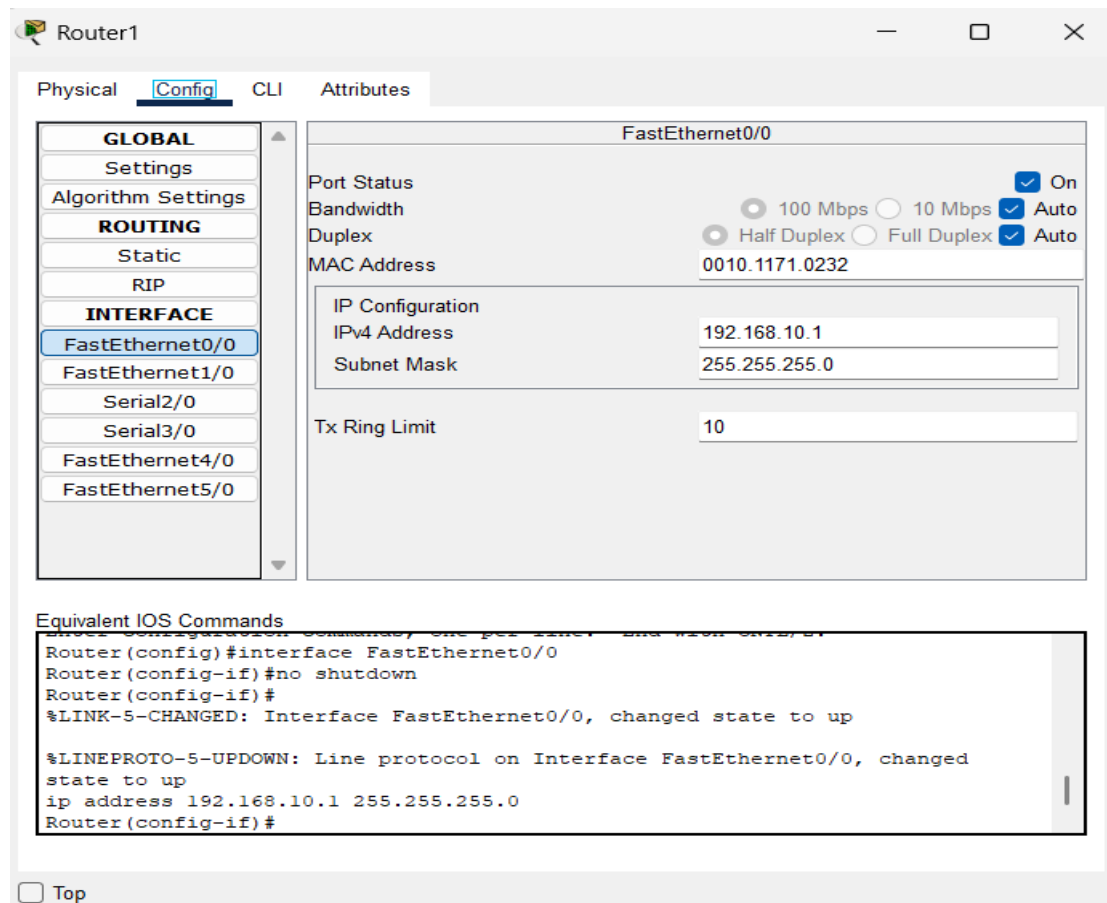


The screenshot shows the 'PC0' window with the 'Desktop' tab selected. The 'IP Configuration' window is open, showing settings for the 'FastEthernet0' interface. The 'Static' option is selected for both IPv4 and IPv6 configurations.

Field	Value
Interface	FastEthernet0
IP Configuration	☐ DHCP ☒ Static
IPv4 Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0
IPv6 Configuration	☐ Automatic ☒ Static
IPv6 Address	
Link Local Address	FE80::204:9AFF:FE52:969C
Default Gateway	
DNS Server	
802.1X	☐ Use 802.1X Security
Authentication	MD5
Username	
Password	

Step 3: Configure Router Interface

- Click the Router → Config → FastEthernet0/0



The screenshot shows the 'Router1' window with the 'Config' tab selected. The 'FastEthernet0/0' interface is selected in the left sidebar. The main configuration area shows the following settings:

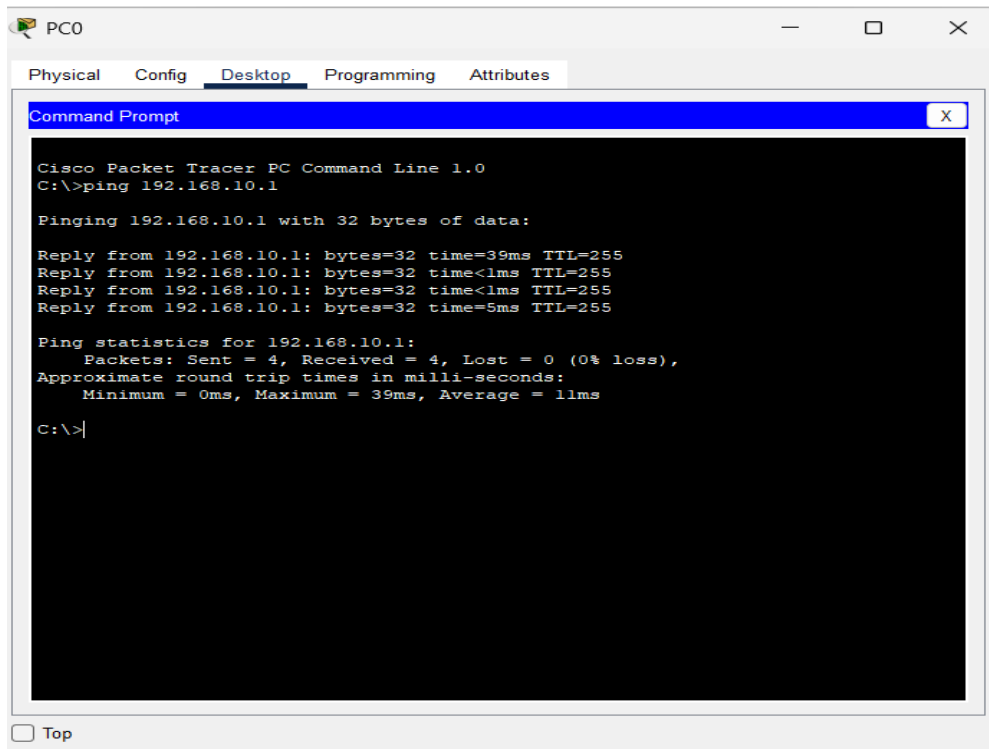
Field	Value
Port Status	☒ On
Bandwidth	☒ 100 Mbps ☐ 10 Mbps
Duplex	☒ Half Duplex ☐ Full Duplex
MAC Address	0010.1171.0232
IP Configuration	☐ DHCP ☒ Static
IPv4 Address	192.168.10.1
Subnet Mask	255.255.255.0
Tx Ring Limit	10

Equivalent IOS Commands:

```
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 192.168.10.1 255.255.255.0
Router(config-if)#
```

Step 4: Test Connectivity

- Open Command Prompt on each PC
- Use the ping command to test connectivity:



The screenshot shows a Cisco Packet Tracer window for PC0. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command prompt displays the output of the 'ping 192.168.10.1' command. The output shows four successful replies with varying times (39ms, <1ms, <1ms, 5ms) and a TTL of 255. The ping statistics indicate 4 packets sent, 4 received, and 0% loss, with an average round trip time of 11ms.

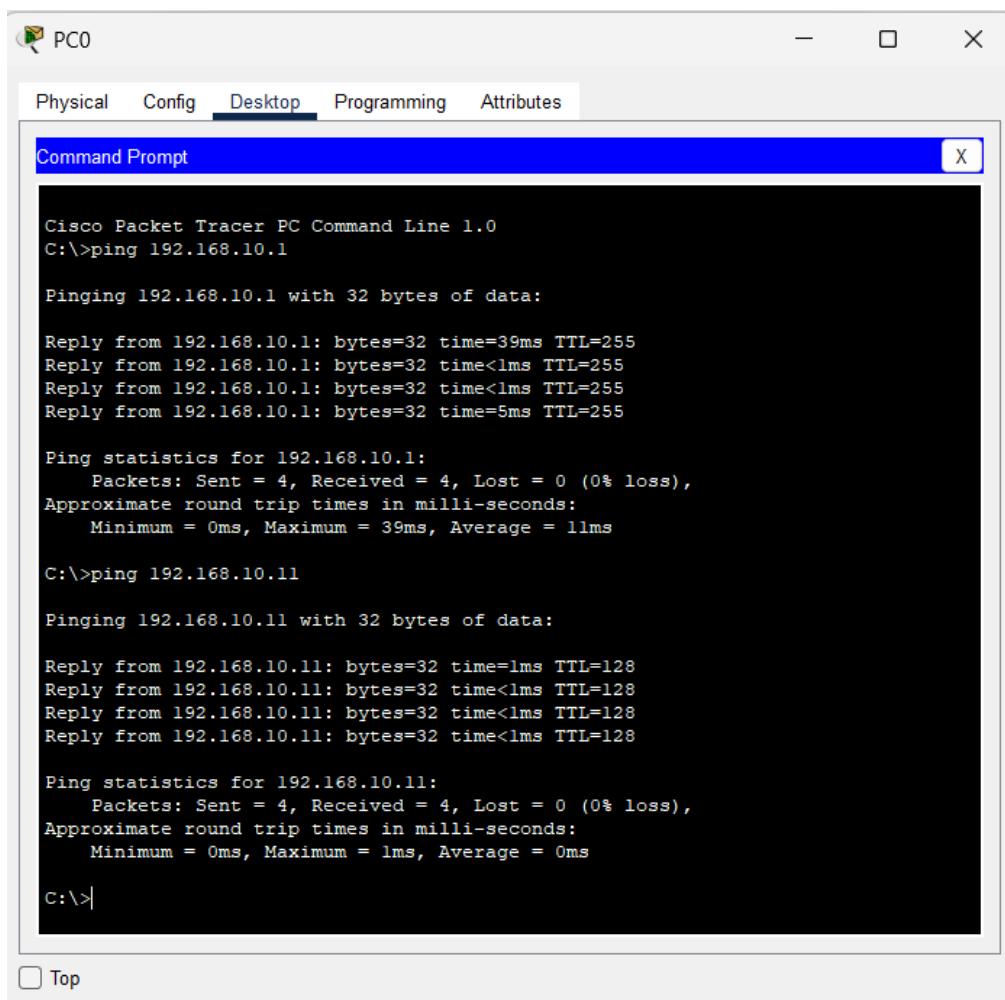
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=39ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=5ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 39ms, Average = 11ms

C:\>
```



The screenshot shows the same Cisco Packet Tracer window for PC0. The 'Command Prompt' window now shows the output of two ping commands. The first command, 'ping 192.168.10.1', is identical to the one in the previous screenshot. The second command, 'ping 192.168.10.11', also shows four successful replies with a time of <1ms and a TTL of 128. The ping statistics for the second command indicate 4 packets sent, 4 received, and 0% loss, with an average round trip time of 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=39ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=5ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 39ms, Average = 11ms

C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Observation:

Each PC successfully sends and receives ping responses to/from:

- The Router (192.168.10.1)
- Other PCs on the same network

Result:

A small LAN with 4 PCs, a switch, and a router was successfully designed and configured. Manual IP addressing was applied, and connectivity was verified using the ping command.

Conclusion:

This lab demonstrated the practical steps of creating a basic LAN, assigning static IP addresses, and testing communication across the network. This is foundational for understanding larger and more complex network topologies.